

02-741 Week 1

Generative AI for Biomedicine

Aidan Jan

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Introduction to Molecular and Cell Biology

We don't quite understand ourselves. In the human body, there are tens of trillions of cells, and dozens of different cell types. (Just neurons have over ten different types.) However, we do not understand how cells differentiate. For example, why a muscle cell originally grew into a muscle cell, or why a neuron originally grew into a neuron. Additionally, we don't know how to change one cell type into another cell type. (e.g., Skin cell $\rightarrow$ Stem cell $\rightarrow$ Neuron.)

One domain of current research is with gene expression. Since all cells have the same DNA, what features (or activation of genes) would cause a stem cell to become a certain cell type? Given some information on which genes are activated, can we predict what cell type the cell is? This is where AI comes in.

Decoding the “Language” of Genomes, Cells, and Tissues

Biology is multi-scale. We can observe gene sequences, cell structure, and cell function, all of which are at different scales. As a result, our models must also be multi-scale.

Central Dogma of Biology

Proteins do most of the work in biology, and are encoded by subsequences of DNA, known as genes.



The Human Genome: the “Blueprint” of Our Body

- 10^{13} different cells in an adult human
- The cell is the basic unit of life
- DNA = linear molecule inside the cell that carries instructions needed throughout the cell's life - long string(s) over a small alphabet
- Alphabet of four nucleotides / bases. (A, C, T, G)

DNA organizes into chromatin, which form into nucleosomes, which condenses into chromosomes. Humans have 22 pairs of chromosomes, plus the two sex chromosomes.

Genomes

- A genome is an organism's complete set of DNA (including its genes)
- In humans, less than 2% of the genome actually encodes for genes

- However, a much larger percentage of the genome is transcribed (miRNAs, lncRNAs, and other non-coding RNAs...)
- ... and a large fraction of the rest of the genome serves as control regions, i.e., these regions are involved in determining when genes are turned on and off depending on the context.

Structure of Genes in Mammalian Cells

- Within coding DNA genes there can be untranslated regions (**introns**)
- The translated parts are termed **exons**. These are the segments of DNA that contain the protein coding information.
- Following transcription, introns are removed and exons are **spliced** to make the **protein**.
- **Alternative splicing** increases the potential number of different proteins, allowing the generation of millions of proteins from a small number of genes.

In recent years (2001 - Present), high throughput DNA sequencing allows for cheap sequencing of human DNA. In 2001, one genome costs over \$100,000,000. Now (2026), it costs less than \$800 per genome. High-throughput DNA sequencing is one of the most disruptive technologies in the past decade.

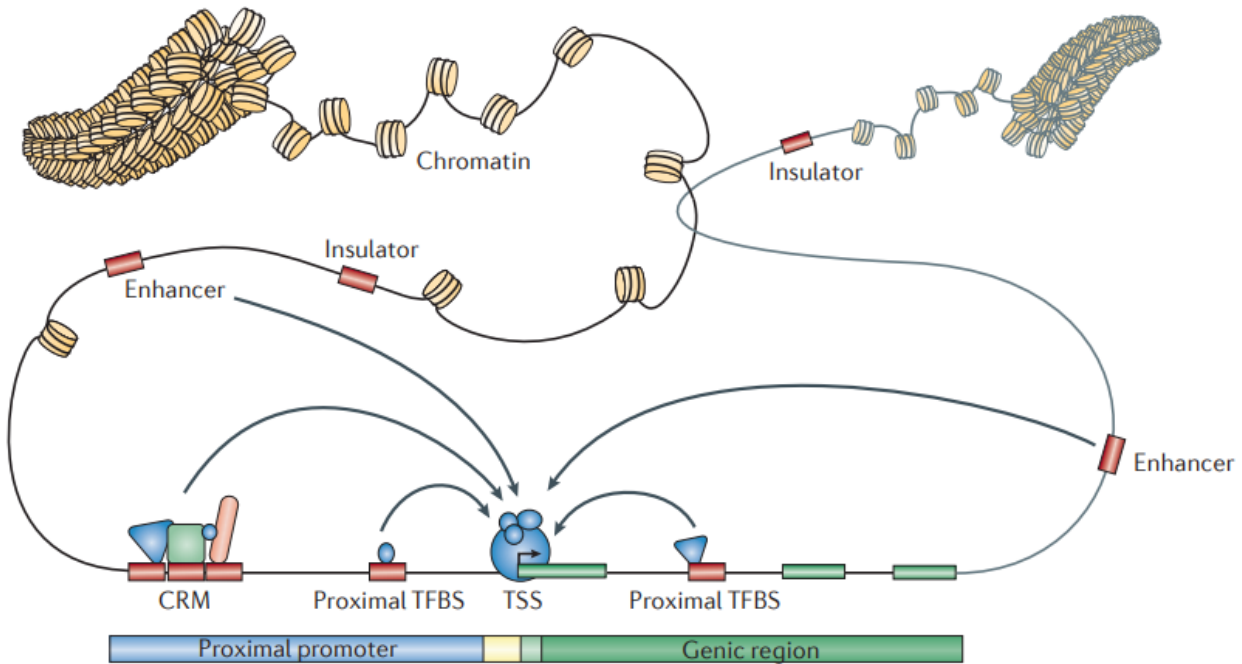
Understanding the Genome

- View from 2000
 - Protein coding regions: 35000-120000
 - Regulatory sequence: Less than protein-coding information
 - Transposons: Junk DNA
 - (All of these assumptions are wrong!)
- We now know...
 - Protein coding regions: 20000. (Fewer than fish!)
 - Regulatory sequences: More regulatory sequences than protein-coding regions
 - Transposons: Make up half of the human genome
 - Gene numbers do not correlate with organism complexity
 - 5% of genes are conserved between different species, but only 1.5% actually codes for proteins

The Spectrum of Genetic Variation

Variation	Rearrangement type	Size range
Single base-pair changes	Single nucleotide polymorphisms (SNPs), point mutations	1 bp
Small insertions / deletions	Binary insertion/deletion events of short sequences (majority <10 bp in size)	1-50 bp
Short tandem repeats	Microsatellites and other simple repeats	1-500 bp
Fine-scale structural variation	Deletions, duplications, tandem repeats, inversions	50 bp to 5 kb
Retroelement insertions	SINEs, LINEs, LTRs, ERVs	300 bp to 10 kb
Intermediate-scale structural variation	Deletions, duplications, tandem repeats, inversions	5 kb to 50 kb
Large-scale structural variation	Deletions, duplications, large tandem repeats, inversions	50 kb to 5 Mb
Chromosomal variation	Euchromatic variants, large cytogenetically visible deletions, duplications, translocations, inversions, and aneuploidy	~5 Mb to entire chromosomes

Transcriptional Regulation



- Transcription start site (TSS)
- Transcription factor binding sites (TFBS)
- Cis-regulatory module (CRM)
- Proximal promoter and distal enhancer

Transcription Factors (TFs) in the Human Genome

- 300 TFs bind to core promoter regions
 - General transcription machinery (e.g., RNA polymerase)
 - Required for transcription of most protein-coding genes
- 1500 TFs bind to other regions in the genome
 - Proximal promoter, enhancer, silencer
 - Regulate a subset of genes

How is Specificity of Binding Achieved?

- Consensus motif
 - Commonly found sequence
 - Shows which nucleotide is most abundant at each position, represented as a Position Weight Matrix (PWM)
 - E.g., GATA1 binding motif: [AT] GATA [AG]
- Motif: A subsequence (substring) that occurs in multiple sequences with a biological importance
- Motifs can be totally constant or have variable elements

- Motifs for regulatory elements
 - Binding sites for proteins
 - Short sequences (5-25 bp)
 - Up to a certain range, e.g., 1000 bp (or farther), from gene
 - Inexactly repeating patterns (challenge!)

From One Cell to Trillions

- Humans have multiple types of cells
- Individual cell types also differentiate into sub-cell types
- Lots more cell types left to discover!

DNA is Only Half the Story

- Variations in DNA alone may not entirely account for variations in phenotypic traits.
- Organisms with identical DNA often exhibit distinct phenotypes
 - Includes plants, insects, mammals, etc.

What is Epigenetics / Epigenomics?

- A mitotically or meiotically heritable state of different gene activity and expression (**phenotype**) that is independent of differences in DNA sequence (**genotype**)
 - - based on Conrad Waddington, 1942
- The sum of the alterations to the chromatin template that collectively establish and propagate different patterns of gene expression (transcription) and silencing from the same genome.
- Epigenetic changes influence the phenotype without altering the genotype.
- While **epigenetics** often refers to the study of single genes or sets of genes, **epigenomics** refers to more global analyses of epigenetic changes across the entire genome.

Epigenetic Mechanisms

- DNA Methylation
 - Normal cells - role in gene expression and chromosome stability
 - Cancer cells - consequences of aberrant hypo- and hyper-methylation
- Histone Modification
 - Normal cells - the histone code
 - Cancer cells - consequences of altered histone modifying enzymes
- Interaction between DNA methylation, histone modifications, and other players such as non-coding RNAs
- Cell and tissue type specificity
- Gene-environment interaction, disease susceptibility

If DNA is like the alphabet, epigenetic marks are like the accents and punctuation. If DNA is like a book, epigenetic marks are like sticky notes.

Chromatin Structures

- Eukaryotic chromatin structure can be viewed as a series of superimposed organizational layers.
- At the root are the DNA sequence and its direct chemical modification by **cytosine methylation**.
- The DNA is folded into **nucleosomes** — the fundamental units of chromatin — that comprise approximately 147 bp of DNA wrapped around a histone octamer.
- The nucleosomal **histones** H2A, H2B, H3 and H4 can be chemically modified and exchanged with variants. The nucleosome positions along with histone variants and modifications make up the primary structure of chromatin.
- Finally, three-dimensional models of chromatin in nuclei are now being developed with increasing precision and propose that there are additional sophisticated layers of genome regulation through **higher-order organization** and **nuclear compartmentalization**.

Higher-Order Genome Organization

- We can now determine a genome's sequence and annotate linear chromatin composition, but our genome is not linear
- Our knowledge of the 3D organization of the genome remains limited
- Need to achieve high spatial and temporal resolution

Sequence Modeling and Generation with Transformer

Can we use GenAI to design molecules with desired functions?

- Medicine
- Vaccines
- Enzymes (Biocatalysts)
- Biosensors (e.g., GFP)
- New materials

Molecule generation is somewhat similar to language generation.

- Both are sequences of discrete tokens. (In language it's words, in molecules, it is atoms or amino acids.)
- Both have discrete structures. (In language it's grammar, in molecules it's chemistry)
- Geometry (Unique for molecules)

For proteins, they can be roughly represented as sequences of amino acids.

- Proteins are building blocks of life
- Responsible for important biological functions
- Built by sequences of amino acids.

– GFP (Green Fluorescent Protein) = VLLPDNHYLSTQSALSKDPNEKRDHMLLEFVTAAGIT

Language Models

A language model is a probabilistic model of discrete sequences, including human languages and biological languages. e.g.,

Probability("Pittsburgh is a city of bridges")

Probability("VLLPDNHYLSTQSALSKDPNEKRDHMLLEFVTAAGIT")

Each next token can be defined as a probability based on previous tokens.

$$P(\text{next word } y_t | \text{Prompt } x, \text{previous words } y_{1:t-1})$$

For example,

- Santa Barbara has very nice _____
 - "beach" = 0.5
 - "weather" = 0.4
 - "snow" = 0.01
- Pittsburgh is a city of _____
 - "bridges" = 0.6
 - "corn" = 0.02

To calculate the probabilities, the probability of each word (or each amino acid) depends on all the previous tokens.

$$\begin{aligned} &P(\text{"Pittsburgh is a city of bridges"}) \\ &= P(\text{"Pittsburgh"}) \cdot P(\text{"is"} | \text{"Pittsburgh"}) \cdot P(\text{"a"} | \text{"Pittsburgh is"}) \cdot P(\text{"city"} | \dots) \\ &\cdot P(\text{"of"} | \dots) \cdot P(\text{"bridges"} | \dots) \end{aligned}$$

Or simply,

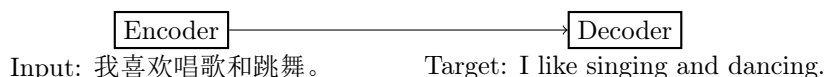
$$P(x_{1:T}) = \prod_{t=1}^T P(x_{t+1} | x_{1:t})$$

We can predict the $P(x_{t+1} | x_{1:t})$ part using neural networks (e.g., CNNs, RNNs)

Types of Language Models

- **Encoder-only Model** (or Masked LM, Non-autoregressive)
 - Examples: BERT, RoBERTa, ESM (for protein)
- **Encoder-decoder**
 - Example: T5
- **Decoder-only** (Autoregressive)
 - Examples: GPT, LLaMA, ProGen (for proteins)

Encoder-Decoder Paradigm

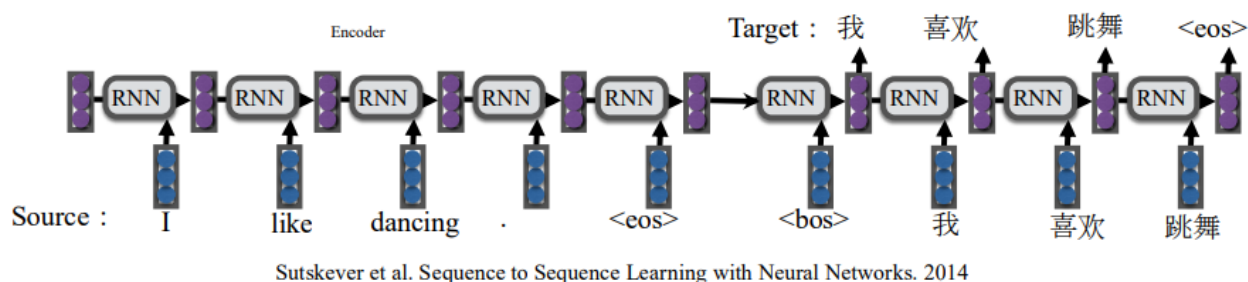


Sequence to Sequence Learning

- Conditional text generation: directly learning a function mapping from source sequence to target sequence

$$p_{\theta}(y|x) = \prod_t p(y_t|x, y_{1:t-1}; \theta)$$

- Previous encoder / decoder: LSTM or GRU



We use the encoder-decoder architecture because it is

- Full context and parallel: use Attention in both encoder and decoder
- Not recurrent $\Rightarrow$ concurrent encoding

Motivation of Attention: Translation between Different Alignment of Words



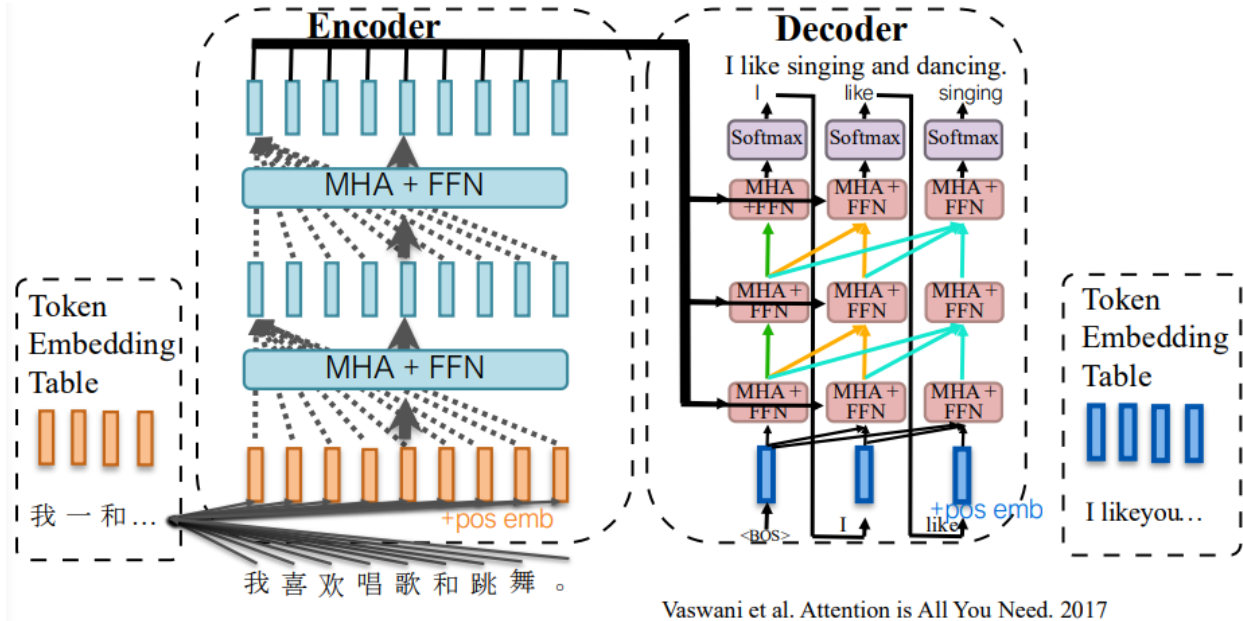
Different languages have different orderings for words.

Transformer-based Models for Biomolecules

- Protein Language Models:
 - ESM, ESM-Fold
 - ProGen
 - AlphaFold (with EvoFormer)
- Genomics
 - DNABERT / DNABERT2

- Evo, Evo2
- ENABERT
- Single Cell: scGPT, scFoundation

Transformers

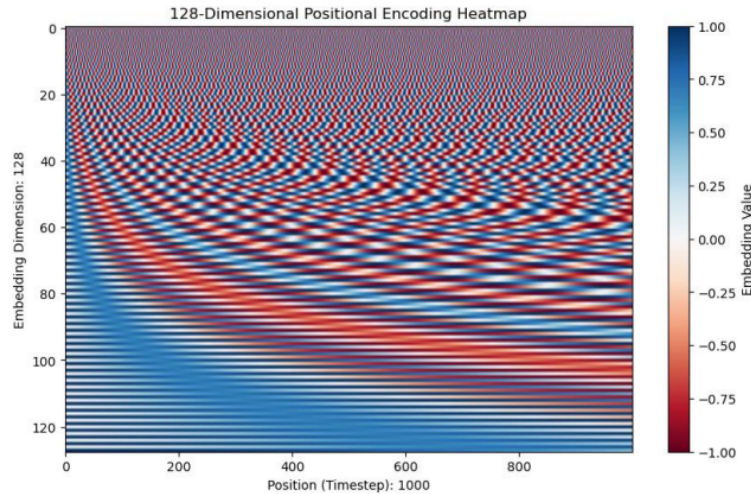


What is Embedding?

- Token Embedding:
 - token index $\rightarrow$ vector
 - Shared (tied) input and output embedding from lookup table
- Positional Embedding
 - to distinguish words in different positions, map position labels to embedding, dimension is same as token embedding
 - For t -th position, i -th dimension, the equation is:

$$PE_{t,2i} = \sin\left(\frac{t}{1000^{2i/d}}\right)$$

$$PE_{t,2i+1} = \cos\left(\frac{t}{1000^{2i/d}}\right)$$

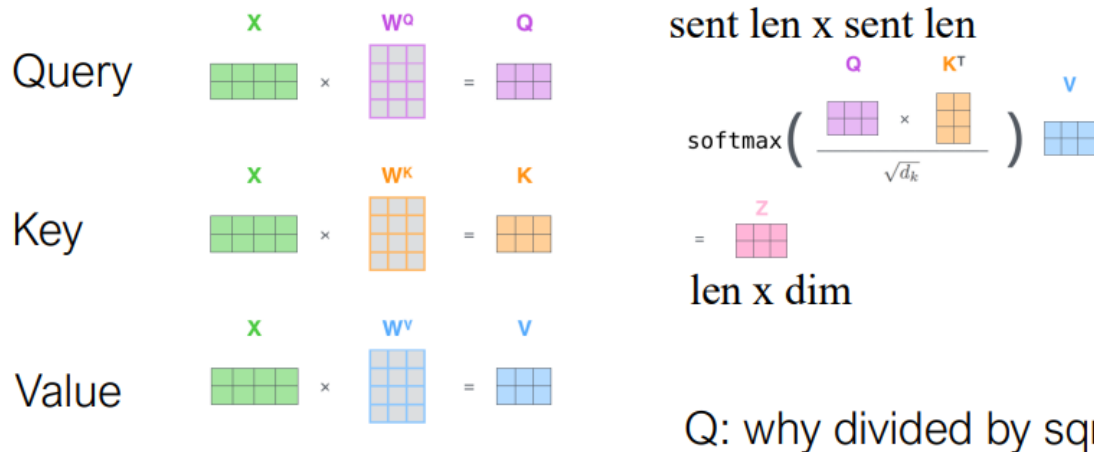


Multi-head Attention

- Instead of one vector for each token, we break it into multiple heads
- Each head performs attention.

$$\begin{aligned}\text{Head}_i &= \text{Attention}(QW_i^Q, KW_i^K, VW_i^V) \text{MultiHead}(Q, K, V) \\ &= \text{Concat}(\text{Head}_1, \text{Head}_2, \dots, \text{Head}_h)W^O\end{aligned}$$

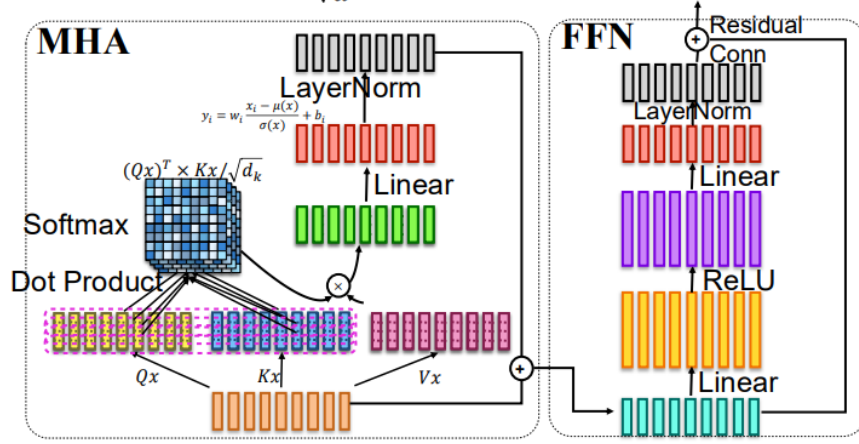
X are input embeddings from previous layer (num of tok * dim)



The $\sqrt{d_k}$ is a regularization coefficient, where d_k is the dimensionality of the attention (number of heads). This coefficient makes it softmax is a range from 0 to 1.

Multihead Attention and FFN

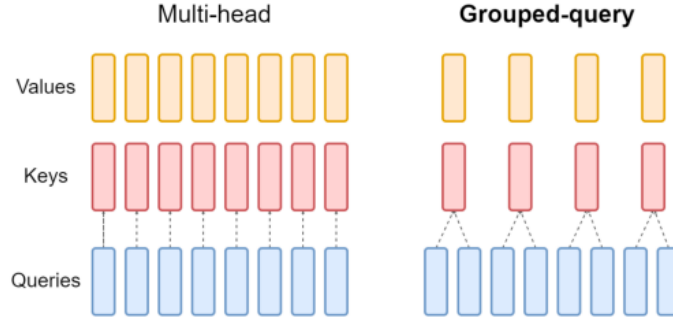
$$\begin{aligned}\text{Attention}(Q, K, V, x) &= \text{Softmax}\left(\frac{(Qx)^T Kx}{\sqrt{d}}\right) \cdot (Vx)^T \\ \text{FFN}(x) &= \max(0, x \cdot W_1 + b_1) \cdot W_2 + b_2\end{aligned}$$



Attention Variant: Grouped-Query Attention (GQA)

$$\text{Head}_i = \text{Attention}(XW_h^Q, XW_{g(h)}^K, XW_{g(h)}^V)$$

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{Head}_1, \text{Head}_2, \dots, \text{Head}_h)W^O$$



GQA: Training Generalized Multi-Query Transformer Models from Multi-Head Checkpoints. Joshua Ainslie, James Lee-Thorp, Michiel de Jong, Yury Zemlyanskiy, Federico Lebrón, Sumit Sanghai. EMNLP 2023.

Grouping the queries can speed up the inference since there are less overall matrix multiplications.

Attention Variant: Multi-head Latent Attention (MLA)

- Keys and values occupy large GPU memory
 $- 2 \cdot \text{len} \cdot n_h \cdot d_h$
- instead of caching d_h vector (for each t_k , head), reduce to a latent space with rank r .

$$Q = X \cdot W_Q^D \cdot W_Q^U$$

$$K = X \cdot W_{KV}^D \cdot W_K^U$$

$$V = X \cdot W_{KV}^D \cdot W_V^U$$

The diagram shows the equations for Q, K, and V. Red arrows indicate the dimensions of the matrices: $d \times r$ for W_Q^D , $r \times (n \cdot d_h)$ for W_Q^U , and $r \times (n \cdot d_h)$ for W_K^U and W_V^U .

- First proposed in Deepseek-V2, used in all modern LLMs

- Key and Value share down-projection, and across all heads

$$\begin{aligned}
& MLA(x)_h \\
&= \text{Softmax} \left(\frac{X \cdot W_Q^D \cdot W_{Q,h}^U \cdot \left(X \cdot W_{KV}^D \cdot W_{K,h}^U \right)^T}{\sqrt{d_h}} \right) \cdot X \cdot W_{KV}^D \cdot W_{V,h}^U \\
&= \text{Softmax} \left(\frac{C_Q \cdot W_{QK,h} \cdot C_{KV}^T}{\sqrt{d_h}} \right) \cdot C_{KV} \cdot W_{V,h}^U
\end{aligned}$$

We can precompute and cache the following:

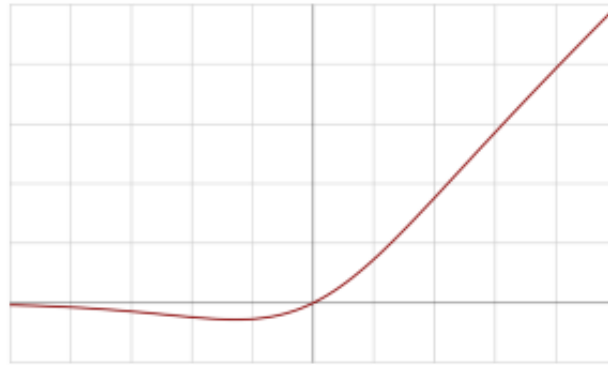
$$\begin{aligned}
W_{QK,h} &= W_{Q,h}^U \cdot (W_{K,h}^U)^T \\
C_Q &= X \cdot W_Q^D \\
C_{KV} &= X \cdot W_{KV}^D
\end{aligned}$$

Decoder Self-Attention Mask

- Encoder uses bidirectional attention
- Decoder uses unidirectional attention
 - Maskout right side before softmax (-inf)

FFN with SwiGLU

The swish activation function looks like



$$\text{swish}(x) = x\sigma(\beta x)$$

$$\sigma(x) = \frac{1}{1 + e^{-x}}$$

FFN with ReLU:

$$\text{FFN}(x) = \max(0, x \cdot W_1 + b_1) \cdot W_2 + b_2$$

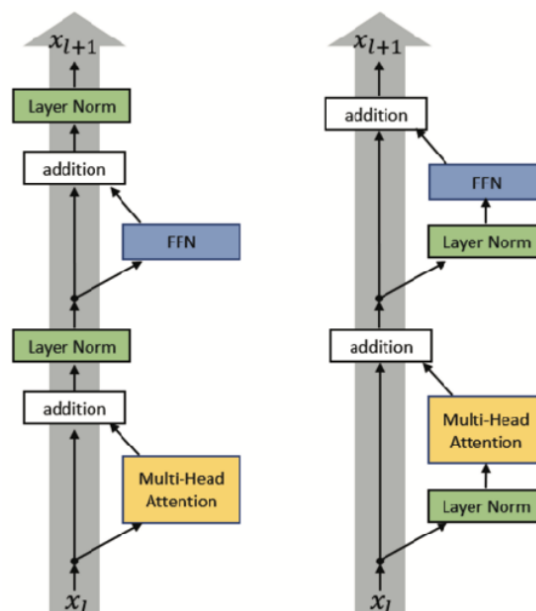
FFN with SwiGLU:

$$\text{FFN}_{\text{SwiGLU}}(x) = (\text{swish}(x \cdot W_1 + b_1) \odot (x \cdot W_2 + b_2)) \cdot W_3 + b_2$$

- FFN with SwiGLU is used in LLaMA and all later LLMs.

Residual Connection and Layer Normalization

- Residual Connection
- Make it zero mean and unit variance within layer
- Post-norm
- Pre-norm



Relative Positional Embedding

- “I have a dream”
- “Today I have a dream”

What are Attention weights for “I” and “dream”? The goal is to make sure the similarity is only dependent on the distance between two embeddings, independent of their absolute positions

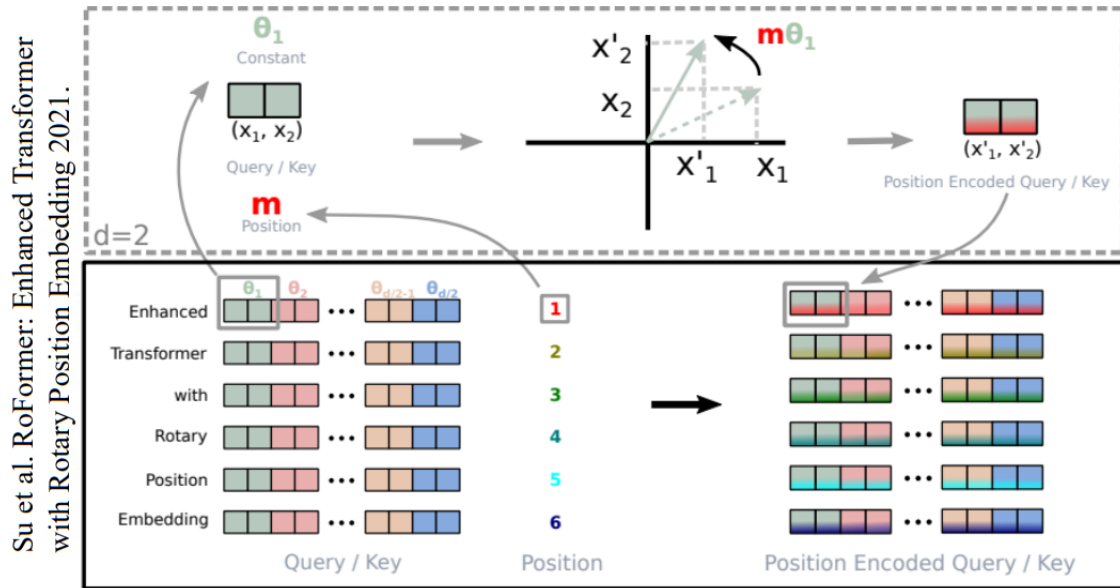
$$Att(x_i, x_j) = Att(x_{i+\Delta}, x_{j+\Delta})$$

Rotary Embedding (RoPE)

Makes the attention weight depending (only) on position distance

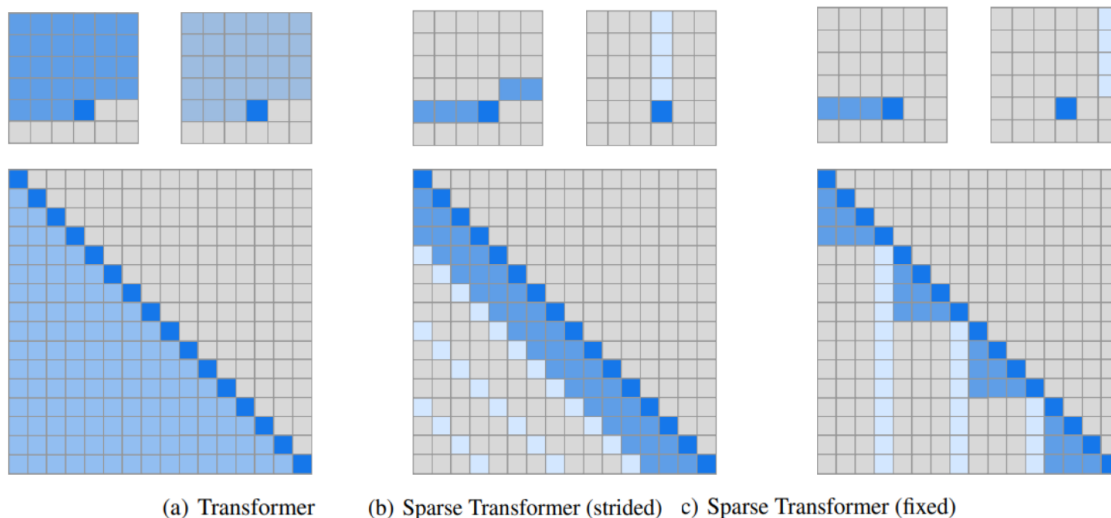
$$f(x_m, m) = \begin{pmatrix} \cos(m\theta) & -\sin(m\theta) \\ \sin(m\theta) & \cos(m\theta) \end{pmatrix} \begin{pmatrix} x_{m,1} \\ x_{m,2} \end{pmatrix}$$

$$f(x_m, m) \cdot f(x_n, n) = x_m^T R_{n-m} x_n$$



Sparse Attention

Alternating dense and locally banded **sparse attention** patterns (GPT3)



Brown, Tom, et al. "Language models are few-shot learners." Advances in neural information processing systems 33 (2020): 1877-1901.

Training Techniques and Performance of Transformer

Training Transformer

$$P(Y|X) = \prod P(y_t|y_{<t}, x)$$

- Training loss: Cross-Entropy

$$\ell = - \sum_n \sum_t \log f_\theta(x_n, y_{n,1}, \dots, y_{n,t-1})$$

- Teacher-forcing during training.
- Pretend to know ground truth for prefix.

Training Transformer for MT

- Dropout
 - Applied to before residual, and to embedding, pos emb.
 - $p = 0.1 \sim 0.3$
- Label smoothing
 - 0.1 probability assigned to non-truth
- Vocabulary
 - En-De: 37K using BPE
 - En-Fr: 32k word-piece (similar to BPE)

Label Smoothing

- Assume $y \in R^n$ is the one-hot encoding of label
- Approximating 0/1 values with softmax is hard

- The smoothed version is

$$y_i = \begin{cases} 1 - \epsilon & \text{if belongs to class } i \\ \epsilon/(n-1) & \text{otherwise} \end{cases}$$

- We commonly use $\epsilon = 0.1$.

Training Technique

- Batch
 - group by approximate sentence length
 - still need shuffling
 - one machine with 8 GPUs (in 2017 paper)
 - base model: 100k steps (12 hours)
 - large model: 300k steps (3.5 days)
- Adam Optimizer
 - increase learning rate during warmup, then decrease
 - $\eta = \frac{1}{\sqrt{d}} \min\left(\frac{1}{\sqrt{d}}, \frac{t}{\sqrt{t_0^3}}\right)$

Adam Optimizer

$$\begin{aligned} m_{t+1} &= \beta_1 m_t + (1 - \beta_1) \nabla \ell(x_t) \\ v_{t+1} &= \beta_2 v_t + (1 - \beta_2) (\nabla \ell(x_t))^2 \end{aligned}$$

Momentum:

$$\hat{m}_{t+1} = \frac{m_{t+1}}{1 - \beta_1^{t+1}}$$

Variance:

$$\hat{v}_{t+1} = \frac{v_{t+1}}{1 - \beta_2^{t+1}}$$

Parameter Update:

$$x_{t+1} = x_t - \frac{\eta}{\sqrt{\hat{v}_{t+1} + \epsilon}} \hat{m}_{t+1}$$

Other Parameter Optimizers

- AdamW (used in LLaMA)
 - Similar to Adap but update with weight decay
- Muon (used in deepseek models)
 - Update 2D weight matrix using fast and orthogonal Newton-Schulz iterations.

Model Average

- A single model obtained by averaging the last 5 checkpoints, which were written at 10-minute intervals (base)
- Decoding length: within source length + 50

Summary

- Key components in Transformer and LLM Architecture
 - Positional Embedding (to distinguish tokens at different positions)
 - Multihead attention
 - Residual connection
 - Layer norm
 - FFN and SwishGLU
 - RoPE
- Training Technique:
 - Dropout
 - Label smoothing
 - Optimization algorithm
- Pretraining: Mask-labeled recovery of random spans and next token prediction
- Multitask supervised fine-tuning with instruction templates
 - T5, InstructGPT